

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

____ show ip route ospf

2

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? ____ Serial 0/0/0 ____

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? ____ 128 ____

Does R2 show up as an OSPF neighbor on R1? ____ No ____

Does R2 show up as an OSPF neighbor on R3? ____ No ____

What does this information tell you?

____ R2 is not forming OSPF adjacency because the interface is configured as passive. ____

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

____ router ospf 1 ____

____ no passive-interface s0/0/1 ____

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? ____ Serial 0/0/1 ____

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

192 ____

Is R2 listed as an OSPF neighbor to R3? ____Yes____ 

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

____To correctly calculate OSPF cost for high-speed interfaces such as Gigabit or faster links. 

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

____OSPF cost is calculated using reference bandwidth divided by interface bandwidth. The total path cost is the sum of all outgoing interface costs along the path. 

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

____1562, Lower bandwidth increases the OSPF cost. 

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

____The path through R2 has a lower OSPF cost than the direct link to R3. OSPF always selects the lowest cost path. 

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?
___ Router ID uniquely identifies each router in the OSPF domain and ensures stable neighbor relationships. ___
4
2. Why is the DR/BDR election process not a concern in this lab?
___ All connections in this lab use point-to-point serial links, which do not require DR/BDR election. ___
4
3. Why would you want to set an OSPF interface to passive?
___ To prevent sending OSPF hello packets to networks where no OSPF neighbors exist. ___
4
4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?
___ The use of areas such as Area 0 (backbone area). ___
4
5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?
___ network 10.0.0.0 0.255.255.255 area 0 ___
4
6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?
___ EIGRP route will be used. ___
4